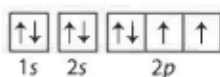


Practice - 1.1

(Excerpt from Chemistry 12, McGraw-Hill Ryerson, 2011 page 193)

1. **K/U** Refer to **Table 3.5**. Draw a table that shows the quantum numbers, electron configuration, and orbital diagram for the electrons in nitrogen.
2. **K/U** A carbon atom has two electrons total in its $2p$ orbitals. If one of these electrons has an electron spin number of $+\frac{1}{2}$, what is the electron spin number of the other electron? Explain your answer in terms of Hund's rule.

3. **T/I** Examine the following orbital diagram for a neutral atom in its ground state.



- a. What element is represented here? How do you know?
 - b. Explain how Hund's rule applies to this orbital diagram.
 - c. Explain how the Pauli exclusion principle applies to this orbital diagram.
4. **T/I** What is wrong with each of the following condensed electron configurations?
 - a. $[\text{Ne}]4s^2$
 - b. $[\text{Kr}]5s^25p^6$
 - c. $[\text{Br}]5s^2$
 5. **T/I** The condensed electron configuration for iodine is $[\text{Kr}]5s^24d^{10}5p^5$. Without using a periodic table, identify the group number to which iodine belongs. Show your reasoning.
 6. **T/I** The condensed electron configuration for germanium is $[\text{Ar}]4s^23d^{10}4p^2$. Without using a periodic table, identify the group number, period number, and orbital block to which germanium belongs. Show your reasoning.
 7. **K/U** Which elements have these valence electron configurations?
 - a. ns^1
 - b. ns^2np^6
 - c. $ns^2(n-1)d^{10}np^4$
 8. **T/I** Use Pauli exclusion principle, Hund's rule, and the aufbau principle to write condensed electron configurations for atoms of the following elements:
 - a. potassium, K
 - b. selenium, Se
 - c. xenon, Xe
 9. **K/U** The ns orbitals usually fill before the $(n-1)d$ orbitals. Why are there exceptions to this rule?
 10. **T/I** Referring only to the periodic table at the back of your textbook, arrange the following sets of elements in order of increasing atomic radius. Briefly explain your reasoning.
 - a. Be, Ca, Ba
 - b. Na, Al, Cl
 - c. Li, Rb, Sn
 11. **C** Your friend Zen is studying for tomorrow's chemistry test and texts you to ask about the difference between Z and Z_{eff} . Write a text to explain, being as brief as possible while including all important information.
 12. **K/U** First ionization energy is an important atomic property.
 - a. What is first ionization energy and how does it relate to an atom's chemical reactivity?
 - b. Briefly summarize and explain the periodic trends in first ionization energy.
 - c. What is the relationship between ionization energy and atomic radius?
 13. **C** Draw diagrams to illustrate why boron, aluminum, oxygen, and sulfur deviate from the periodic trends in ionization energy.
 14. **A** The extremely reactive Group 1 metals are commonly stored in oil.
 - a. What is the purpose of the oil?
 - b. Suggest at least one safety precaution a chemist would need to take when working with Group 1 metals.
 - c. Why are Group 1 metals so reactive?

Further Practice:

5. Which one of the following electron configurations represents an electron in an excited state?

a. $[\text{Ar}]4s^23d^{10}4p^3$
 b. $[\text{Ar}]4s^23d^{10}4p^6$
 c. $[\text{Ar}]4s^23d^{10}4p^2$
 d. $[\text{Ar}]4s^23d^{10}4p^4$
 e. $[\text{Ar}]4s^23d^{10}4p^35s^1$

6. Which of the following elements could have the following valence orbital diagram?



- a. carbon
 b. nitrogen
 c. boron
 d. sulfur
 e. neon
7. First ionization energy is
- a. the amount of energy an electron must gain to escape from a neutral atom.
 b. the amount of energy an electron must lose to escape from a neutral atom.
 c. the amount of energy an atom gains when it first becomes an ion.
 d. Two of these are correct.
 e. All of these are correct.
8. Electron affinity refers to
- a. the energy change accompanying the addition of 1 mol of gaseous atoms or ions.
 b. the attraction of the nucleus for electrons.
 c. the attraction of electrons for one another.
 d. the energy change of electrons of noble gas elements.
 e. none of the above.
9. In which period can you start to use condensed electron configurations?
- a. 1
 b. 2
 c. 3
 d. 4
 e. 5
10. Atomic radius decreases
- a. when the effective nuclear charge decreases.
 b. when the inner core decreases.
 c. down a group.
 d. across a period.
 e. in all of the above situations.

11. Which property changes in the same way that the atomic radius does in the periodic table?

a. electron affinity
 b. atomic number
 c. ionization energy
 d. They all change in the same way.
 e. None of them change in the same way.

12. Which two sets of quantum numbers describe electrons in the same sublevel?

a. i, ii
 b. iii, v
 c. ii, iii
 d. iv, v
 e. ii, iv

	n	l	m_l	m_s
i.	2	1	0	$-\frac{1}{2}$
ii.	2	1	-1	$+\frac{1}{2}$
iii.	3	1	-1	$+\frac{1}{2}$
iv.	2	2	-1	$+\frac{1}{2}$
v.	3	2	-1	$+\frac{1}{2}$

13. When comparing the properties of carbon atoms and oxygen atoms, which statement is correct?

a. Carbon and oxygen have the same spin quantum number, m_s .
 b. Carbon and oxygen have the same magnetic quantum number, m_l .
 c. Carbon and oxygen have the same principal quantum number, n .
 d. Carbon has a greater orbital-shape quantum number, l , than oxygen.
 e. Oxygen has a greater orbital-shape quantum number, l , than carbon.

14. Helium behaves like a noble gas because

a. its valence shell is completely filled.
 b. its outermost orbital is completely filled.
 c. its outermost energy level is completely filled.
 d. Two of these are correct.
 e. All of these are correct.

21. Explain how periodic trends in ionization energy and electron affinity explain why atoms of elements in Group 1 and Group 2 tend to bond with other elements by forming positive ions in ionic compounds.
22. A nitrogen atom has a total of three electrons in its $2p$ orbitals. Are any of these electrons paired? Explain how you know.
23. The quantum mechanical model provides information about the atom that is certain and uncertain. Which information is certain? Which is uncertain?
24. What information do the quantum numbers provide about an orbital?
25. What two things does the principal quantum number, n , describe about an orbital and the electron in it?
26. Why does it make sense that an electron can be found farther away from the nucleus in an orbital of a higher energy level than in the same type of orbital in a lower energy level?
27. What is the Pauli exclusion principle, and how is it related to the spin quantum number, m_s ?
28. What is Hund's rule and its significance?
29. Why does it make sense that electron configurations represent atoms in their ground state and not their many excited states?
30. What are the general trends for atomic radius down a group and across a period in the periodic table? Account for these trends.
32. Without consulting a periodic table, determine to which period, group, and block the element with the electron configuration $[\text{Xe}]6s^25d^{10}$ would belong.
33. Write the complete electron configuration for the element in Period 5 and Group 15.
34. List the $3s$, $5s$, $2p$, $4p$, $3d$, and $5d$ orbitals in order, from the lowest energy to the highest energy.
35. List all possible values of m_l for electrons in the element having the electron configuration $[\text{Ar}]4s^23d^5$.
36. Use a basic periodic table, which does not contain electron configurations, to determine which element is defined by $[\text{Ar}]4s^23d^8$.
37. Use a basic periodic table, which does not contain electron configurations, to determine which element is defined by $[\text{Ne}]2s^22p^1$.
38. Use a basic periodic table, which does not contain electron configurations, to determine which element is defined by $[\text{Kr}]5s^24d^{10}5p^4$.
39. If the last electron to be added to an atom was the seventh electron in the third energy level, what element would that be?
40. If the last electron to be added to an atom was the 44th electron in the atom, what element would that be?
41. If the last electron to be added to an atom was the sixth electron in the second energy level, what element would that be?
42. If the last electron to be added to an atom was the second electron in the sixth energy level, what element would that be?
43. Some chemists do not include the d electrons in the condensed electron configurations for large p -block elements. For example, they would write the condensed electron configuration for tin as $[\text{Kr}]5s^22p^2$. Why do you think this might be a valid way of writing the electron configurations?
44. Use Z_{eff} to predict which atom should be larger between boron and fluorine.
45. Use Z_{eff} to predict which atom should be larger between magnesium and silicon.
46. Use Z_{eff} to predict which atom should be larger between calcium and selenium.
48. **T/I** Referring only to a periodic table, arrange the following sets of elements in order of increasing first ionization energy. Briefly explain your reasoning.
 - a. Na, Si, Ar
 - b. Mg, Ca, Ba
 - c. He, Li, Be
49. **T/I** Consider the electron configuration: $1s^22s^22p^4$.
 - a. Assume the electron configuration represents a neutral atom in its ground state. What element does it represent? Explain how you know.
 - b. What information does this electron configuration notation provide?
 - c. What information does this electron configuration notation *not* provide?

Text (Chemistry 12, Mcgraw-Hill Ryerson 2011)

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